

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Kanmani L/192511414 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: _____

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.
10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

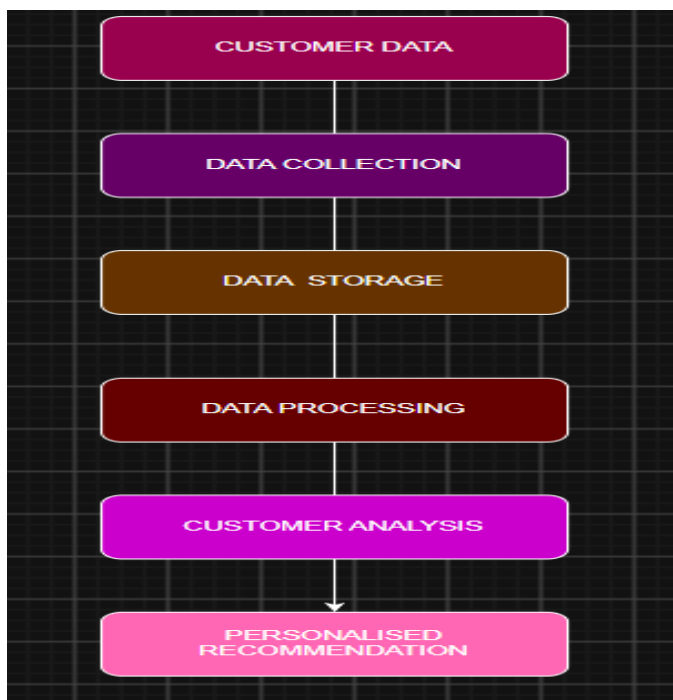
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.

Answer:

A multinational retail organization collects customer data from different sources such as online websites, mobile applications, physical stores, CRM systems, loyalty programs, and social media. If this data is fragmented, it becomes difficult to understand customer behavior and provide personalized services.

- A unified big data architecture can integrate all these data sources into a common platform. Real-time data can be collected using Apache Kafka and stored in a cloud-based data lake such as Amazon S3 or Azure Data Lake. Apache Spark can process both historical and real-time data.
- A centralized customer profile can then combine purchase history, browsing behavior, preferences, and demographic information.
- Machine learning models can analyze this unified data to generate personalized product recommendations. Business intelligence tools such as Power BI can display real-time customer analytics to managers.
- Architecture:
Online/Offline Data → Kafka → Data Lake → Spark → Customer Profile → ML Recommendation → Dashboard
- This architecture provides real-time insights, better personalization, improved customer experience, and scalable data processing.



2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.

Answer:

Scientific research produces huge amounts of data from fields such as genomics, astronomy, climate studies, and scientific simulations. Processing such data using a centralized system can be slow because one computer has limited processing power, memory, and storage.

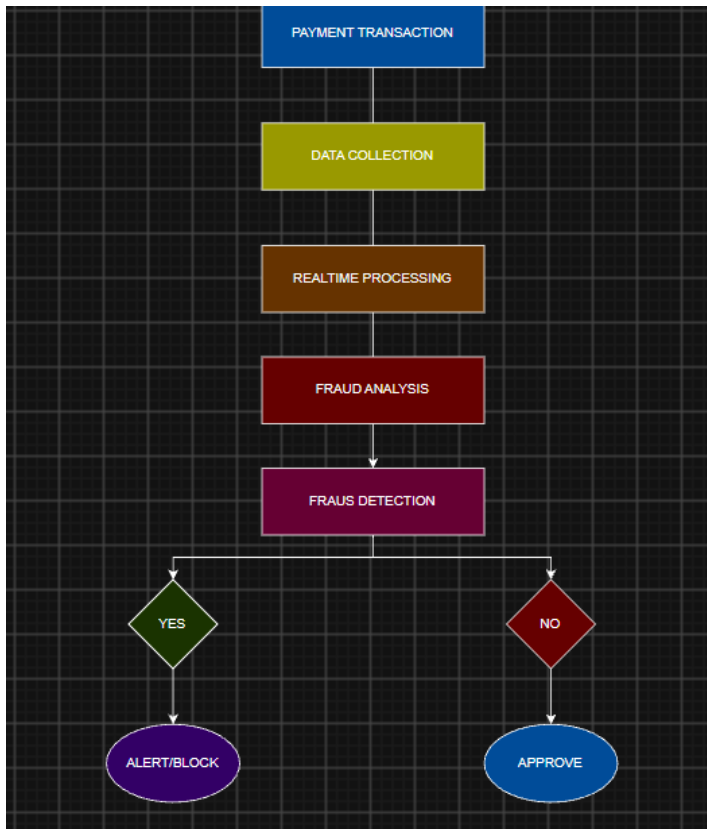
- Distributed computing solves this problem by dividing large datasets and computational tasks among multiple computers called nodes. Technologies such as Hadoop and Apache Spark support distributed and parallel processing.
- In parallel processing, several nodes perform different tasks simultaneously. For example, a large scientific dataset can be divided into smaller partitions, with each partition processed by a different node. This significantly reduces execution time.
- Distributed computing also provides horizontal scalability, since additional nodes can be added when the workload increases. It provides fault tolerance because the system can recover from individual node failures.
- Therefore, distributed computing is more effective than centralized processing for large scientific datasets because it provides faster computation, scalability, fault tolerance, and better resource utilization.

3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.

Answer:

Digital payment platforms generate a continuous stream of transactions. Fraud detection must identify suspicious transactions quickly before financial losses occur.

- A big data fraud detection framework can use Apache Kafka to collect real-time transaction data. Spark Streaming or Apache Flink can process the incoming data. Important features such as transaction amount, location, device information, transaction frequency, time, and spending patterns can be extracted.
- Machine learning algorithms such as Random Forest, XGBoost, and Isolation Forest can analyze these features and calculate a risk score. If a transaction is identified as highly suspicious, the system can generate an alert, request additional verification, or block the transaction.
- Framework:
Transactions → Kafka → Stream Processing → Feature Extraction → ML Model → Risk Score → Alert/Block
- Streaming processing enables immediate fraud detection, while predictive models can identify complex patterns that traditional rule-based methods may miss. Hence, big data analytics improves fraud detection accuracy, response time, and payment security.



4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.

Answer:

Multimedia data includes images, audio, and video, which creates several challenges in a big data environment. These files have large storage requirements and different formats. Processing them also requires high computational power and network bandwidth.

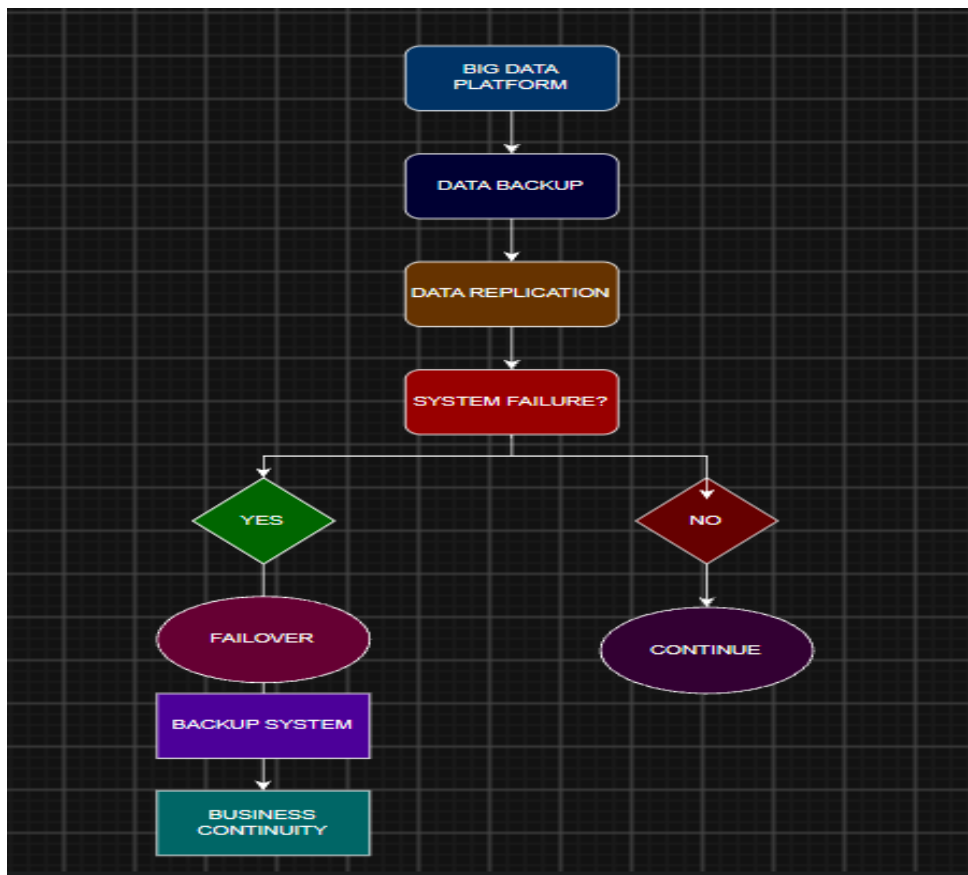
- A suitable solution is to use distributed cloud object storage such as Amazon S3, Azure Blob Storage, or Google Cloud Storage. HDFS can also be used in Hadoop-based environments
- Metadata such as file name, format, duration, and user information can be stored in NoSQL databases such as MongoDB.
- Technologies such as Apache Spark can perform large-scale processing and analysis. Compression can reduce storage requirements, while transcoding can convert multimedia into suitable formats. Content Delivery Networks (CDNs) can be used to efficiently deliver videos and other media to users.
- Therefore, distributed storage, cloud object storage, NoSQL databases, compression, and parallel processing provide an efficient and scalable solution for managing multimedia big data.

5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.

Answer:

A cloud-based big data platform may face failures due to hardware problems, network failures, cyberattacks, software errors, or natural disasters. A disaster recovery and business continuity strategy ensures that data and services remain available or can be restored quickly.

- The organization should maintain redundant servers and storage across multiple availability zones or geographical regions. Important data should be replicated to another location, and regular backups should be maintained.
- Automated failover can redirect users to a backup server when the primary server fails. Load balancing can distribute requests among multiple servers, preventing a single point of failure. Organizations should also define Recovery Time Objective (RTO), which specifies the maximum acceptable downtime, and Recovery Point Objective (RPO), which specifies the maximum acceptable data loss.
- Regular recovery testing should be performed to ensure that the strategy works correctly.
- Thus, redundancy, replication, backups, multi-region deployment, and automated failover provide operational resilience and help minimize downtime and data loss during failures.



6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.

Answer:

Data quality is very important in a large-scale analytics platform because business decisions and machine learning models depend on accurate data. Common data quality problems include inconsistency, duplication, and missing values.

Inconsistent data occurs when the same information is stored in different formats, such as different date or address formats. Duplicate data may cause the same customer or transaction to be counted multiple times. Missing values can reduce the accuracy of statistical analysis and machine learning predictions. These problems can result in incorrect reports, unreliable insights, and poor business decisions.

To solve these problems, organizations should implement data governance strategies. Data cleansing can remove duplicates and standardize formats, while missing values can be handled using appropriate imputation techniques. Master Data Management (MDM) can maintain consistent customer and product information.

Data owners and data stewards should be responsible for maintaining data quality. Data validation rules, metadata management, data catalogs, access controls, and regular quality monitoring should also be implemented.

Thus, effective data governance ensures that data is accurate, complete, consistent, secure, and reliable, improving the quality of analytics and decision-making

7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.

Answer:

An online streaming service generates huge amounts of user behavior data, including watch history, search history, ratings, likes, viewing duration, and skipped content. This data can be analyzed to provide personalized recommendations.

- User activities can be collected in real time using Apache Kafka and stored in a cloud-based data lake. Apache Spark can process large volumes of historical and real-time data using distributed computing. Machine learning techniques such as collaborative filtering, content-based filtering, or hybrid recommendation can then be used to identify content that users are likely to enjoy.
- For example, if a user frequently watches action movies, the recommendation system can suggest similar movies based on their viewing history and the behavior of users with similar preferences.
- Architecture:
User Activity → Kafka → Data Lake → Spark → ML Recommendation Model → Personalized Content

- Distributed processing allows the system to handle millions of users and large datasets efficiently. Continuous analysis of user behavior improves recommendation accuracy.

Therefore, a scalable recommendation system can increase personalization, user engagement, viewing time, satisfaction, and customer retention.



8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.

Answer:

Cloud computing provides the infrastructure required to store, process, and analyze massive amounts of big data. Platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud provide scalable storage, computing, databases, and analytics services.

- One major advantage is scalability. Organizations can increase or decrease computing resources according to their workload. Cloud computing also provides flexibility because resources can be accessed on demand. The pay-as-you-go model reduces the need for large investments in physical hardware.
- However, cloud computing also has limitations. Poor resource management can result in unexpected costs. Organizations may face vendor lock-in, network dependency, security risks, privacy concerns,

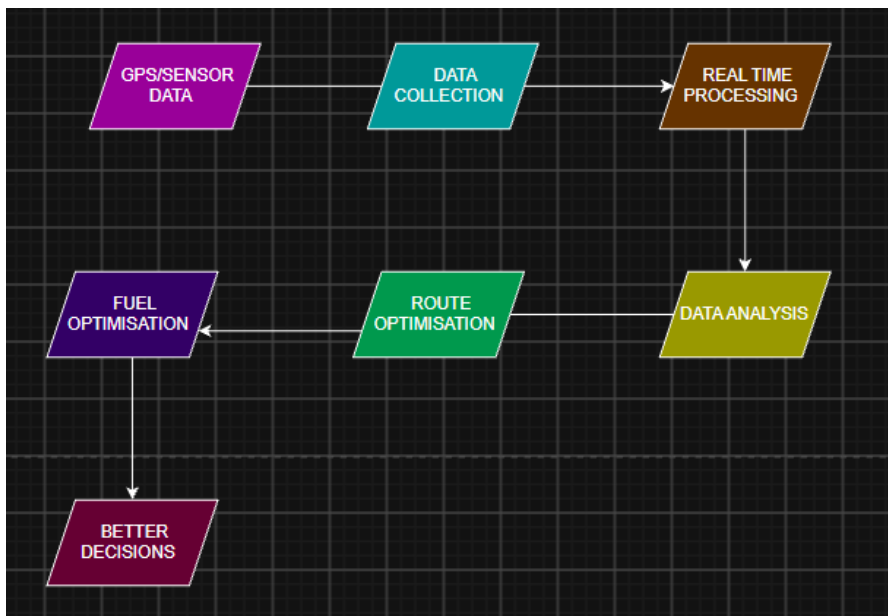
and regulatory issues. Moving very large datasets through networks can also be expensive and time-consuming.

- Organizations can address these challenges using cost monitoring, encryption, identity and access management, security policies, and proper data governance.
 - Hence, cloud computing provides scalability, flexibility, and efficient resource management for big data, but security, cost, and compliance must be carefully managed.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

Answer:

Transportation and logistics systems generate large amounts of real-time data from GPS devices, vehicle sensors, traffic systems, weather services, fuel monitoring systems, and delivery applications. Analyzing this data helps organizations optimize transportation operations.

- IoT devices can continuously send data to Apache Kafka, which acts as a real-time data ingestion system. Apache Spark Streaming or Apache Flink can process the incoming data. Analytics models can analyze vehicle locations, traffic conditions, fuel consumption, and delivery schedules.
- Real-time route optimization can identify the fastest and most efficient routes based on current traffic conditions. Predictive analytics can identify vehicles that require maintenance, while fuel monitoring can identify inefficient driving patterns.
- Architecture:
GPS/IoT Sensors → Kafka → Stream Processing → Real-Time Analytics → Route Optimization → Dashboard/Alerts
- Big data technologies help reduce travel time, fuel consumption, vehicle downtime, and delivery delays. Therefore, real-time analytics improves route optimization, fuel efficiency, fleet management, and operational decision-making.



10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Answer:

Big data provides valuable insights, but collecting and analyzing large amounts of personal information creates several ethical and legal challenges. Major concerns include privacy violations, unauthorized data collection, lack of consent, data breaches, and misuse of personal information.

- Machine learning algorithms can also produce bias and discrimination if the data used to train them is biased. Lack of transparency can make it difficult for users to understand how their data is being used or how automated decisions are made. Organizations must also comply with applicable data protection and privacy regulations.
- To ensure responsible data usage, organizations should establish policies for informed consent, data minimization, encryption, access control, anonymization, secure storage, and data retention. Regular security audits should be performed, and users should be informed about how their data is collected and used.
- Organizations should also test AI models for bias and fairness and maintain transparency in automated decision-making.
- Thus, responsible big data management requires organizations to balance business benefits with privacy, security, fairness, legal compliance, and user trust.